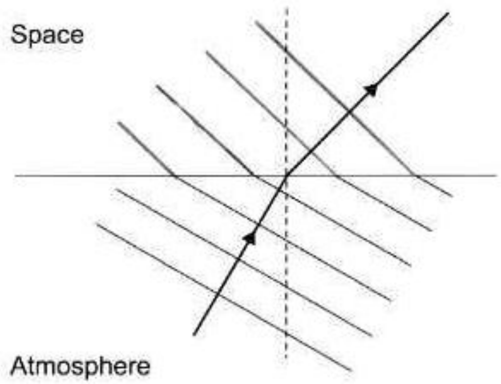
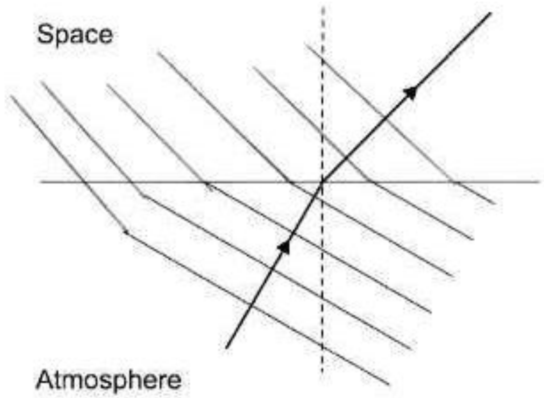
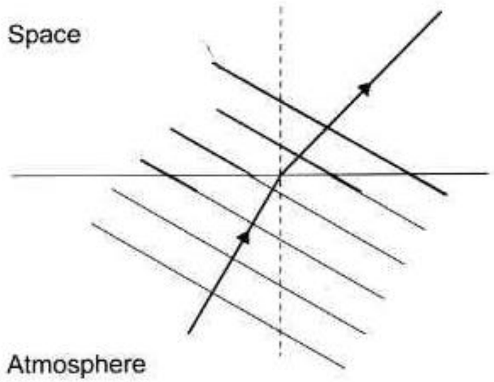
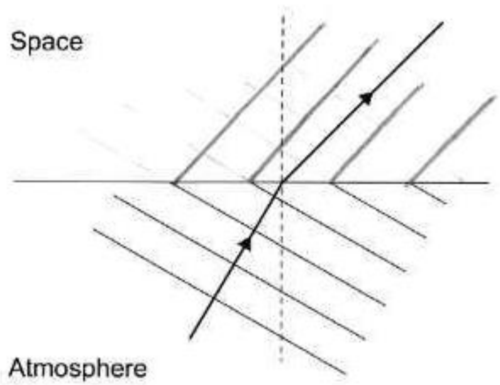
Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
2.	(a)	(i)		Wavelength = 6 [cm]		1		1	1	
		(ii)		Conversion to m = 0.06 m (1) Substitution and manipulation: $f = \frac{3 \times 10^8}{0.06\text{ ecf}}$ (1) $f = 5 \times 10^9$ [Hz] (1) Award 2 marks If no conversion, answer of $f = 5 \times 10^7$ [Hz] Award 2 marks for answers of 5×10^n where n is not 9		3		3	3	
	(b)			Wavefronts perpendicular to direction (1) Longer wavelengths (1) 3 or more wavefronts joined at boundary (1) To earn any credit wavefronts must be travelling towards the top right corner **Examples**  **3 marks**		3		3		

				 **3 marks** **Only consider additions to the 4 wavefronts in contact with the boundary**  **1 mark**						

				 **No marks**						
	(c)	(i)		Satellite orbits in the same time / 24 hours (1) that it takes the earth to spin once (1) Accept for 1 mark they both orbit in 24 hours	2			2		
		(ii)		Total distance = 4 $\times$ (1) [$3.6 \times 10^7 = 14.4 \times 10^7$ [m]] Substitution: $3 \times 10^8 = \frac{14.4 \times 10^7 \textbf{ ecf}}{\text{time}}$ (1) Time = 0.48 [s] (1) Award 2 marks for answers of 48×10^n **or** 0.24 [s] Award 1 mark for answers of 24×10^n **or** 0.12 [s]	 1	1 1		3	3	
		(iii)		Distance travelled would be the same **or** both have 4 steps **or** still use 2 satellites [so disagree]			1	1		
				Question 2 total	**3**	**9**	**1**	**13**	**7**	**0**